

SMART AGRICULTURE USING IoT

1.1 INTRODUCTION

Agriculture plays a major role in supporting human life by providing food, raw materials, and employment. However, farmers face several difficulties, including limited water availability, changing weather, soil problems, shortage of labour, and crop-related issues. Conventional farming methods may not always provide accurate information about field conditions.

Internet of Things (IoT) technology provides a way to improve agricultural practices through continuous sensing and automated control. Smart Agriculture combines sensors, a controller, communication technology, and software to collect and analyse information from the field. Parameters such as soil moisture, temperature, humidity, and water level can be monitored in real time.

Using the collected information, the system can automatically control irrigation and help farmers make better decisions. This approach can improve resource utilisation, reduce unnecessary effort, and support more efficient and sustainable farming.

1.2 NEED FOR SMART AGRICULTURE

The demand for agricultural production is increasing, while important resources such as water and fertile land are limited. Manual irrigation can lead to excess water usage because it is not always based on the actual moisture condition of the soil. Weather variations can also influence crop development, and a shortage of farm labour can make regular monitoring difficult.

Smart Agriculture addresses these concerns by continuously observing field conditions and responding according to predefined requirements. Automated irrigation can supply water when the soil becomes dry and stop the pump when adequate moisture is restored. Remote monitoring also enables farmers to obtain field information without having to check every parameter manually.

1.3 SMART AGRICULTURE USING IoT

An IoT-based Smart Agriculture system connects field sensors with a microcontroller and an Internet-enabled monitoring platform. Sensors measure important agricultural parameters and send their readings to the controller. The controller evaluates the readings and transfers relevant information to a cloud service or mobile/web application.

When a monitored value crosses a selected limit, the system can perform an appropriate action. For example, if soil moisture falls below the required level, the controller can activate a relay and start the water pump. Once sufficient moisture is detected, the pump can be stopped. This creates a continuous monitoring and control cycle that reduces manual intervention.

1.4 COMPONENTS AND TECHNOLOGIES USED

Soil Moisture Sensor: Detects the moisture condition of the soil and provides the main input for irrigation control.

Temperature and Humidity Sensor: Measures the surrounding temperature and humidity of the agricultural area.

Water-Level Sensor: Checks the available water level in the storage tank.

ESP32 / Arduino: Receives sensor readings, processes the data, and controls the connected devices.

Relay Module: Provides switching control for the water pump.

Water Pump: Delivers water to the crop area whenever irrigation is required.

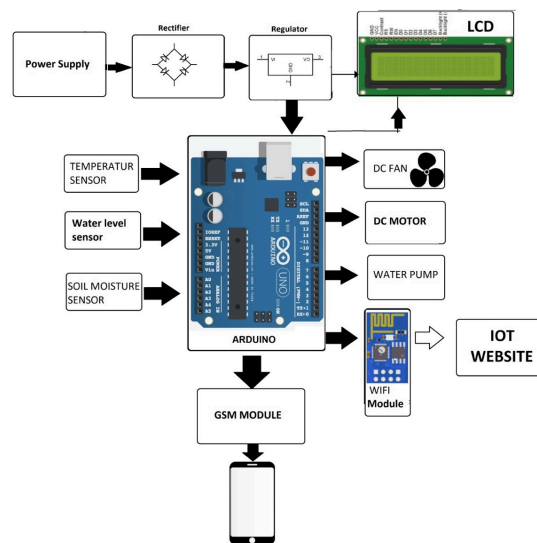
Wi-Fi / Internet: Transfers collected information between the controller and the online platform.

Cloud / Mobile Application: Displays field information so that the farmer can monitor conditions remotely.

1.5 WORKING OF IoT BASED SMART AGRICULTURE

1. **Field Sensing:** Sensors collect soil moisture, temperature, humidity, and water-level information.
2. **Controller Processing:** The ESP32 or Arduino receives the sensor readings and evaluates them.
3. **Data Communication:** The controller sends the relevant readings through Wi-Fi/Internet.
4. **Remote Monitoring:** The information can be viewed through a cloud platform or mobile/web application.
5. **Condition Evaluation:** The controller compares the sensor values with predefined operating limits.
6. **Pump Control:** When soil moisture is insufficient, the controller energises the relay and starts the pump.
7. **Water Supply:** The pump irrigates the crop area until the required moisture condition is achieved.
8. **Automatic Repetition:** The sensors continue measuring the field, allowing the cycle to repeat automatically.

1.6 BLOCK DIAGRAM



Overall flow: Field Sensors → Sensor Interface → ESP32/Arduino → Decision Unit → Relay → Water Pump, with Wi-Fi/Internet providing remote monitoring through the cloud/mobile application.

1.7 APPLICATION

Smart Irrigation – Supplies water according to the measured soil condition.

Soil Condition Monitoring – Provides regular information about the field environment.

Weather and Microclimate Monitoring – Tracks temperature and humidity around crops.

Crop Health Support – Continuous field data can help farmers notice abnormal conditions earlier.

Smart Greenhouses – Sensors and controllers can be used to maintain suitable greenhouse conditions.

Pest and Disease Management – IoT data can support systems designed for early identification of crop problems.

Fertilizer Management – Field information can assist in applying inputs according to crop requirements.

1.8 ADVANTAGES

Reduces unnecessary water usage through condition-based irrigation.

Decreases the need for continuous manual monitoring.

Provides field information in near real time.

Supports better planning of irrigation and other agricultural activities.

Enables remote observation through a mobile or web interface.

Can improve the efficiency of agricultural resource utilisation.

Supports more sustainable farming practices.

1.9 CHALLENGES AND LIMITATIONS

Implementation of an IoT-based agricultural system requires sensors, controllers, communication equipment, and other hardware, which can increase the initial investment. The system also depends on reliable power and Internet connectivity for remote monitoring. Sensors need suitable placement, periodic maintenance, and calibration to maintain dependable measurements. In addition, users may require basic technical knowledge to operate and maintain the system. These factors should be considered before large-scale deployment.

1.10 FUTURE SCOPE

The system can be extended by combining IoT with Artificial Intelligence (AI) and Machine Learning (ML). Historical and real-time agricultural data can be analysed to support prediction of irrigation needs, crop problems, and weather-related conditions.

Future improvements may include:

Drone-assisted crop and field observation.

Robotic systems for planting, spraying, and harvesting.

AI-supported identification of crop diseases.

Automatic and data-driven fertilizer application.

Improved weather forecasting and field alerts.

Integrated farms with a higher level of automated operation.

1.11 CONCLUSION

IoT-based Smart Agriculture provides a practical method for monitoring field conditions and automating selected farming activities. By combining sensors, a microcontroller, Internet connectivity, and irrigation control, the system can provide useful information and respond automatically to changing soil and environmental conditions.

Although installation cost, connectivity, maintenance, and technical requirements can affect implementation, further development in AI, robotics, drones, and IoT can increase the capabilities of smart farming systems. Overall, this approach can contribute to efficient resource usage, reduced manual effort, and more sustainable agricultural management.